typically contain detailed information about structural elements, mechanical systems, electrical networks, plumbing infrastructure, and architectural features, along with their material properties, performance specifications, and maintenance requirements. This integrated approach enables architects, engineers, and facility managers to analyze building performance, simulate different scenarios, and coordinate complex construction and renovation projects with unprecedented accuracy.

The *industry foundation classes (IFC)* are an ISO standard (https://www.iso.org/standard/84123.html) that serves as the primary open file format for exchanging BIM data. IFC files contain not only the geometric representation of building elements but also their semantic meaning—a wall object understands that it's a wall, knows its thermal properties, structural capacity, and relationships to adjacent spaces and systems.

You can navigate to https://3dviewer.net/#model=assets/models/haus.ifc to view a simple BIM model defined as an IFC file as shown in figure 2.10, showing both the geometric model and the properties associated with the selected wall.

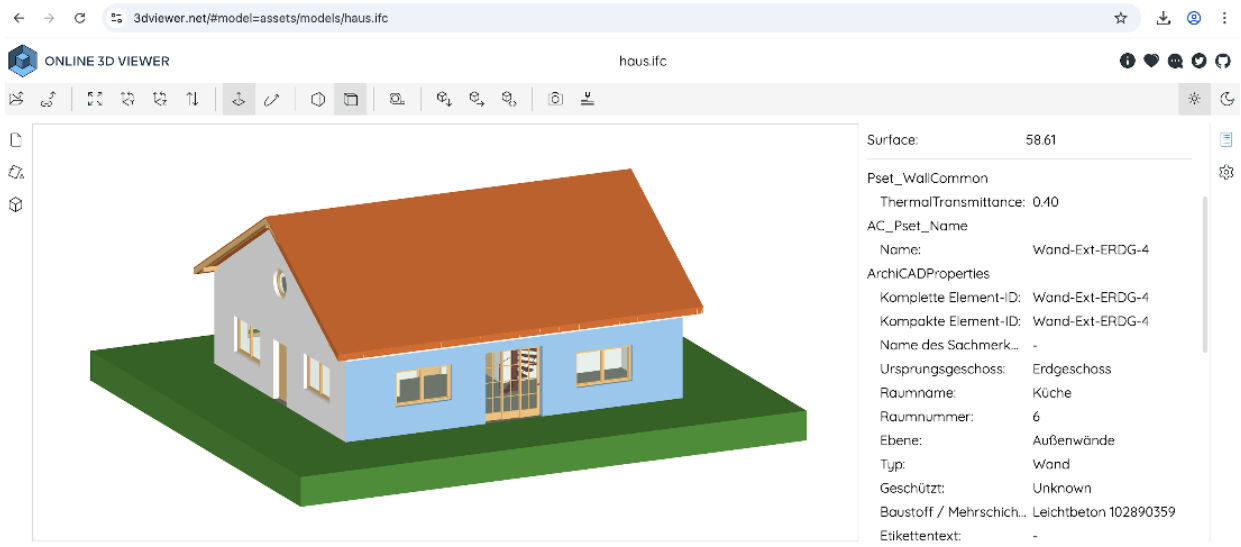


Figure 2.10 An example of a BIM model in IFC showing the 3D model and additional information stored within it. © Viktor Kovacs. Data licensed under MIT.